

OPTIMISTIC INITIAL VALUES - NUMERICAL SOLUTION

Complete Step-by-Step Walkthrough

PROBLEM SCENARIO

Setup

- **5-armed bandit** with 20 total plays
 - **True reward distributions:**
 - Arm 0: $p=0.30$, $\text{mean}=3.0$ | Arm 1: $p=0.50$, $\text{mean}=5.0$
 - Arm 2: $p=0.60$, $\text{mean}=6.0$ **OPTIMAL** | Arm 3: $p=0.20$, $\text{mean}=2.0$
 - Arm 4: $p=0.45$, $\text{mean}=4.5$
 - **Key Strategy:** Initialize $Q(a) = 5.0$ (optimistic!) for all arms
 - **Selection:** Pure **greedy** (no parameter!)
 - **Mechanism:** Optimism-reality discrepancy drives exploration automatically
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ALGORITHM DESCRIPTION

Optimistic Initialization Principle

1. Start: $Q(a) = Q_{\text{init}} = 5.0$ for ALL arms (optimistic guess)
2. First try of each arm: reward will likely be < 5.0 (disappointment)
3. $Q(a)$ drops below 5.0 after first try
4. Untried arms still at $Q(a) = 5.0$ (look more promising!)
5. Natural incentive to try untried arms (exploration emerges!)
6. Once arm tried and estimated, use greedy (exploit best)

KEY: Exploration via DISCREPANCY, not randomness!

Why Optimism Works

All arms start optimistic at $Q = 5.0$:

- Try any arm → Likely get < 5.0 → Q drops
 - Other untried arms still at 5.0 → Still look good
 - Try another arm → Same disappointment → Q drops
 - Continue until all tried → Then exploit best
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COMPLETE NUMERICAL WALKTHROUGH

INITIAL STATE (Play 0)

Arm	Q(a)	N(a)	Assessment
0	5.0	0	"Looks promising!"
1	5.0	0	"Looks promising!"
2	5.0	0	"Looks promising!"
3	5.0	0	"Looks promising!"
4	5.0	0	"Looks promising!"

Greedy selection: All tied at 5.0 $\rightarrow$ Pick Arm 0

PLAY 1

Greedy Selection: All Q=5.0 $\rightarrow$ Pick Arm 0

Reward: Arm 0, p=0.30, draw=0.42 $\rightarrow$ **Reward: 0**

Update:

$Q_{old}(0) = 5.0, N_{old}(0) = 0$
 $error = 0 - 5.0 = -5.0$ (DISAPPOINTMENT!)
 $N_{new}(0) = 1$
 $Q_{new}(0) = (5.0 \times 0 + 0) / 1 = 0.0$

State: Q = [0.0, 5.0, 5.0, 5.0, 5.0]

Key Insight: Arm 0 is now UNATTRACTIVE (Q=0 < 5.0). Other arms still at 5.0!

PLAY 2

Greedy Selection: Q = [0, 5, 5, 5, 5] $\rightarrow$ Pick Arm 1 (first with Q=5.0)

Reward: Arm 1, p=0.50, draw=0.45 $\rightarrow$ **Reward: 10** (LUCKY!)

Update:

$Q_{old}(1) = 5.0$
 $error = 10 - 5.0 = 5.0$ (BEAT expectations!)
 $Q_{new}(1) = (5.0 \times 0 + 10) / 1 = 10.0$

State: Q = [0.0, 10.0, 5.0, 5.0, 5.0]

Note: Arm 1 beat optimism! Q=10 > 5.0 now

PLAY 3

Greedy Selection: $\text{argmax } Q = \text{Arm 1 } (Q=10.0)$

Reward: Arm 1, $p=0.50$, draw=0.61 $\rightarrow$ **Reward: 0**

Update:

$Q_{\text{old}}(1) = 10.0$, $N_{\text{old}}(1) = 1$
 $\text{error} = 0 - 10.0 = -10.0$ (disappointed)
 $Q_{\text{new}}(1) = (10.0 \times 1 + 0) / 2 = 5.0$

State: $Q = [0.0, 5.0, 5.0, 5.0, 5.0]$

Key: Arm 1 back to 5.0! Now all untried arms tied at 5.0!

PLAY 4

Greedy Selection: $Q = [0, 5, 5, 5, 5] \rightarrow$ Pick Arm 2

Reward: Arm 2, $p=0.60$, draw=0.35 $\rightarrow$ **Reward: 10**

Update:

$Q_{\text{new}}(2) = 10.0$ (first try, lucky!)

State: $Q = [0.0, 5.0, 10.0, 5.0, 5.0]$

Critical: Found Arm 2 (THE OPTIMAL ARM)!

PLAYS 5-20 SUMMARY

Play	Arm	Reward	Q Before	N Before	Q After	Note
5	3	0	5.0	0	0.0	Explore
6	2	10	10.0	1	10.0	Stay
7	2	0	10.0	2	5.0	Variance
8	2	10	5.0	3	6.67	Best rises
9	1	10	5.0	1	7.5	Explore
10	2	10	6.67	4	7.5	Converging
11	2	10	7.5	5	8.0	Still rising
12	2	0	8.0	6	6.67	Variance
13	4	10	5.0	0	10.0	Last explore
14-20	2	Mixed	Varying	7-13	Varies	Exploitation

By play 14+: Arm 2's Q is clearly highest $\rightarrow$ Greedy exploits it

Final after 20 plays:

Q = [0.0, 7.5, 8.5, 0.0, 10.0]

N = [1, 2, 12, 1, 2]

Total Reward: ~95

COMPARISON: OPTIMISTIC vs GREEDY

Metric	Greedy	Optimistic
Total Reward	95	~95
Arms Tried	1-2	4-5
Found Optimal	NO	YES
Plays on Optimal	0	12
Discovery Method	None	Optimism
100+ plays?	Stuck ~500	~600-650

Optimistic advantage shows in LONG run: Discovers best arm early!

HOW OPTIMISM DRIVES EXPLORATION

The Mechanism

All optimistic (Q=5.0):

Pull Arm 0 → Get 0 → Disappointed (Q drops to 0)

Pull Arm 1 → Get 10 → Excited (Q=10)

Pull Arm 1 → Get 0 → Disappointed (Q=5.0)

Now Arm 2 at 5.0 looks as good as Arm 1

Pull Arm 2 → Get 10 → NEW BEST ARM FOUND!

Continue exploiting Arm 2

No explicit needed! Exploration emerges naturally!

Why Better Than -Greedy

-Greedy: Random exploration wastes tries on obviously bad arms

Optimistic: Exploration is INTELLIGENT - tries promising untested arms

PARAMETER SELECTION: Q_init

Effect of Different Initial Values

Q_init = 0: Pure greedy, no exploration (Q=0 is realistic)

Q_init = 2: Mild optimism (arms need lucky pulls to beat it)

Q_init = 5: Moderate optimism ← USED HERE (beats expectations)
Q_init = 10: Strong optimism (takes many tries to estimate)
Q_init = 20: Extreme optimism (overexplores, late convergence)

Rule: Q_init max expected reward works well

Why Q_init = 5 for This Problem

Max possible reward = 10
Arm means = [3.0, 5.0, 6.0, 2.0, 4.5]

Q_init = 5 is:

- Below max reward (10) → realistic
 - Above most arm means → creates optimism
 - Near middle → balances discovery vs convergence
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KEY INSIGHTS

Advantages of Optimistic Init

1. **No explicit parameter** - cleaner, more elegant
2. **Intelligent exploration** - tries untested arms, not random
3. **Natural decay** - exploration fades as reality emerges
4. **Fast discovery** - finds good arms relatively quickly

Disadvantages

1. **Requires tuning Q_init** - need to know reward scale
2. **Not guaranteed exploration** - could still get stuck (rare)
3. **Exploration is gradual** - might miss arms given few plays
4. **Advantage fades** - in long run, all methods converge

When to Use Optimistic

- **Unknown domain** - don't know reward range
 - **Prefer elegant solutions** - avoid parameter tuning
 - **Discovery phase matters** - need to find good arms early
 - **Recommend for academic/clean implementations**
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FORMULAS USED

Optimistic Initialization

$$Q(a) = Q_{init} \text{ for all } a \quad (Q_{init} = 5.0)$$

Greedy Action Selection

$$a^* = \arg \max_a Q(a)$$

Q-Value Update

$$Q_{new}(a) = \frac{Q_{old}(a) \times N_{old}(a) + R_t}{N_{new}(a)}$$

SUMMARY

What Happened

1. **Plays 1-2:** Tried Arm 0 (disappointed), Arm 1 (lucky)
2. **Play 3:** Arm 1 dropped, Arm 2 looked promising
3. **Play 4:** Found Arm 2 (optimal!) by exploration
4. **Plays 5-20:** Mostly exploited Arm 2
5. **Result:** Total reward ~95, found optimal arm

Why It Worked

Optimism created natural exploration: Arms beaten by expectations got less selection, while untested arms at $Q=5$ stayed attractive. This drove exploration without explicit randomness.

Key Insight

Initialization matters! Starting optimistic ($Q_{init}=5$) is equivalent to saying: *“I believe all arms are good until proven otherwise.”* This skeptical optimism drives the algorithm to test various arms before committing.

Lesson for Real-World

Use optimistic initialization when: - Building recommendation systems - Testing new features - A/B testing different approaches - You want elegant, parameter-light exploration